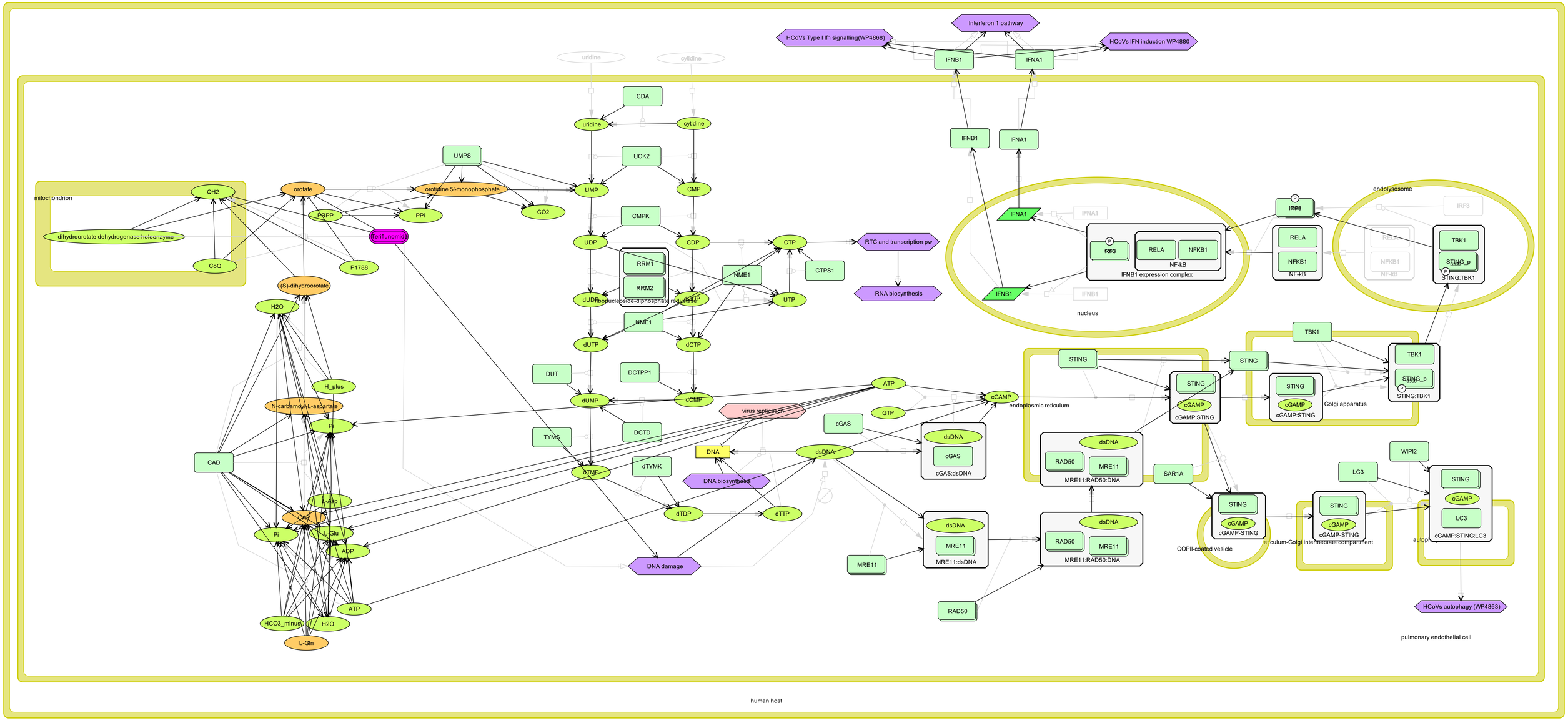

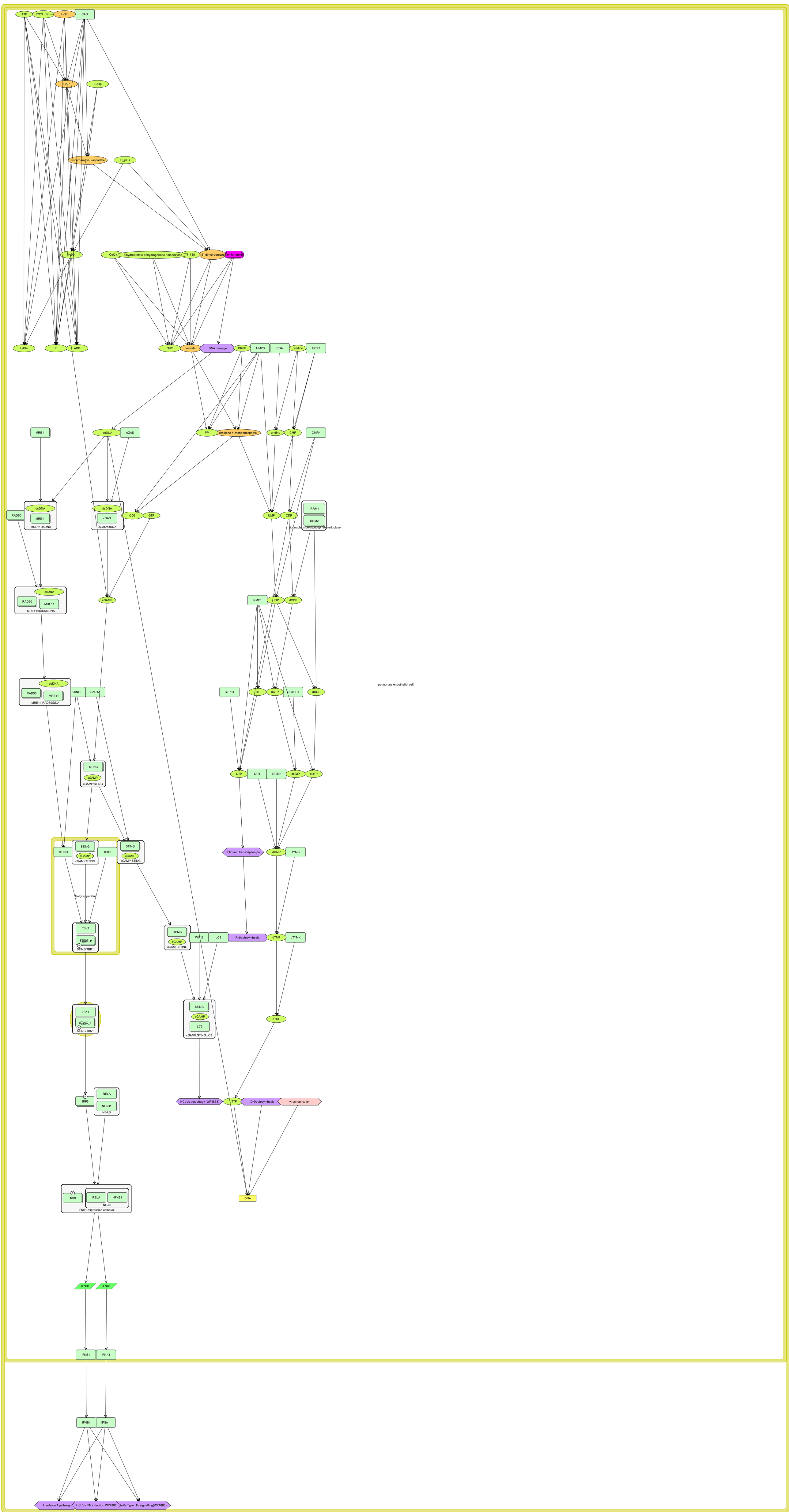

Interferon 1 pathway
HCoVs Type I Ifn signalling(WP4868)
HCoVs IFN induction WP4880
IFNB1
IFNA1
uridine
cytidine
CDA
uridine
cytidine
UMPS
UCK2
orotate
orotidine 5'-monophosphate
UMP
CMP
QH2
mitochondrion
PRPP
PPi
CO2
dihydroorotate dehydrogenase holoenzyme
Teriflunomide
CoQ
P1788
CMPK
UDP
CDP
CTP
RTC and transcription pw
RNA biosynthesis
CTPS1
RRM1
RRM2
ribonucleoside-diphosphate reductase
NME1
dUDP
dCDP
UTP
(S)-dihydroorotate
H2O
NME1
dUTP
dCTP
IFNB1
IFNA1
IFNA1
IFNA1
IFNB1
IFNB1
P
IRF3
RELA
NFKB1
NF-kB
IFNB1 expression complex
nucleus
P
IRF3
RELA
NFKB1
NF-kB
endolysosome
IRF3
RELA
NFKB1
NF-kB
TBK1
STING_p
S366
P
STING:TBK1
DUT
DCTPP1
dUMP
dCMP
H_plus
N-carbamoyl-L-aspartate
Pi
ATP
GTP
cGAMP
endoplasmic reticulum
virus replication
cGAS
DCTD
TYMS
CAD
DNA
dsDNA
dsDNA
cGAS
cGAS:dsDNA
dTMP
dTYMK
DNA biosynthesis
L-Asp
CAP
L-Glu
Pi
ADP
dTDP
dTTP
DNA damage
dsDNA
MRE11
MRE11:dsDNA
MRE11
dsDNA
RAD50
MRE11
MRE11:RAD50:DNA
dsDNA
RAD50
MRE11
MRE11:RAD50:DNA
RAD50
ATP
HCO3_minus
H2O
L-Gln
STING
STING
cGAMP
cGAMP:STING
STING
STING
cGAMP
cGAMP:STING
Golgi apparatus
TBK1
TBK1
STING_p
S366
P
STING:TBK1
SAR1A
STING
cGAMP
cGAMP-STING
COPII-coated vesicle
STING
cGAMP
cGAMP-STING
endoplasmic reticulum-Golgi intermediate compartment
LC3
WIPI2
STING
cGAMP
LC3
cGAMP:STING:LC3
autophagosome
HCoVs autophagy (WP4863)
pulmonary endothelial cell
human host

ATP
HCO3_minus
L-Gln
CAD
CAP
L-Asp
N-carbamoyl-L-aspartate
H_plus
H2O
CoQ
dihydroorotate dehydrogenase holoenzyme
31788
(S)-dihydroorotate
Teriflunomide
L-Glu
Pi
ADP
QH2
orotate
DNA damage
PRPP
UMPS
CDA
cytidine
UCK2
MRE11
dsDNA
cGAS
PPi
orotidine 5'-monophosphate
uridine
CMP
CMPK
RAD50
dsDNA
MRE11
MRE11:dsDNA
dsDNA
cGAS
cGAS:dsDNA
CO2
GTP
UMP
CDP
RRM1
RRM2
ribonucleoside-diphosphate reductase
dsDNA
RAD50
MRE11
MRE11:RAD50:DNA
cGAMP
NME1
UDP
dCDP
dsDNA
RAD50
MRE11
MRE11:RAD50:DNA
STING
SAR1A
CTPS1
UTP
dCTP
DCTPP1
dUDP
pulmonary endothelial cell
STING
cGAMP
cGAMP:STING
CTP
DUT
DCTD
dCMP
dUTP
STING
STING
cGAMP
cGAMP:STING
TBK1
STING
cGAMP
cGAMP-STING
RTC and transcription pw
dUMP
TYMS
Golgi apparatus
TBK1
STING_p
S366
P
STING:TBK1
STING
cGAMP
cGAMP-STING
WIPI2
LC3
RNA biosynthesis
dTMP
dTYMK
TBK1
STING_p
S366
P
STING:TBK1
STING
cGAMP
LC3
cGAMP:STING:LC3
dTDP
P
IRF3
RELA
NFKB1
NF-kB
HCoVs autophagy (WP4863)
dTTP
DNA biosynthesis
virus replication
DNA
P
IRF3
RELA
NFKB1
NF-kB
IFNB1 expression complex
IFNB1
IFNA1
IFNB1
IFNA1
IFNB1
IFNA1
Interferon 1 pathway
HCoVs IFN induction WP4880
CoVs Type I Ifn signalling(WP4868)